



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to AMMAN AGENCIES MULANUR

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

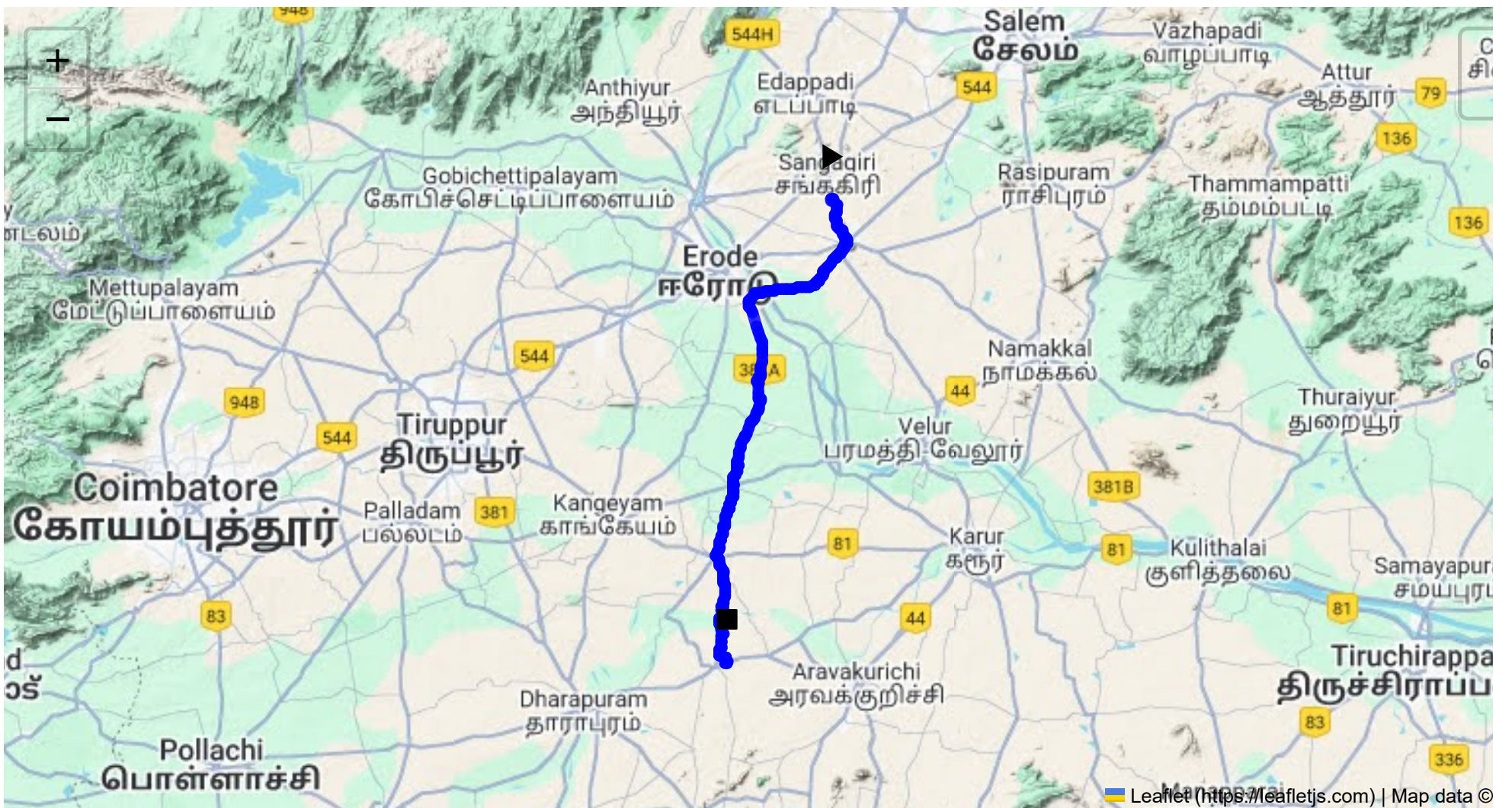
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 87.16 km**  
**Estimated Duration: 2.1 hours**  
**Adjusted Duration (Heavy Vehicle): 2.6 hours**  
**Start: (11.4381, 77.8734)**  
**End: (10.799796, 77.725254)**



## Welcome to the Journey Risk Management Study

### Route Analysis Report

#### 1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to QPXG+W4 Moventhar Nagar, via 8Q58+55P Erode Outer Ring Rd, features several key segments through the state of Tamil Nadu. It initially involves traveling through minor



roads in Sangagiri before linking to the Erode Outer Ring Road, eventually transitioning onto highways and arterial roads towards Mooventhar Nagar.

## 2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate with high temperatures and humidity. The region sees substantial rainfall during the monsoon seasons (June to September and October to November). This can lead to potential hazards like water-logged roads or reduced visibility. Temporary road closures due to flooding or landslides in monsoon periods should be anticipated.

## 3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (8:00-10:00 AM) and evening (5:00-7:00 PM) see significant traffic, especially around urban centers like Erode.
- **Congestion-Prone Areas:** Erode Outer Ring Road can get congested during peak hours. The entry and exit points leading to and from major highways may experience bottlenecks.

## 4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Varies across the route. Urban sections near Erode are generally well-maintained, while certain rural sections may have potholes or uneven surfaces.
- **Infrastructure:** Adequate signage and street lighting in most parts, but less so in rural areas.

## 5. Suggestions for Alternative Routes for Emergencies

In emergencies, bypass traffic congestion by taking internal roads and smaller village paths for a short distance until reconnecting with the main route, primarily around urban areas like Erode.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu mandates permits for hazardous materials transport, requiring adherence to specific routes and times (usually restricted to non-peak hours). Loading and unloading should happen in designated areas.

## 7. Overview of Historical Incidents

In recent years, the region has experienced a few vehicular accidents involving trucks, often due to human error or adverse weather conditions. However, major incidents involving hazardous materials are rare but underline the need for compliance with safety guidelines.

## 8. Environmental Considerations and Sensitive Areas

Route passes near agricultural lands and small residential settlements. The outskirts of Erode are particularly sensitive due to their ecological and social significance—drivers should be cautious of

pollution and noise levels.

## 9. Analysis of Communication Coverage

Communication coverage is generally good near urban areas but can be spotty in rural and less populated zones. Identifying potential network dead zones is crucial for maintaining regular updates on position and condition.

## 10. Estimated Emergency Response Times

- Urban Areas:** Approximately 30-45 minutes response time in Erode due to proximity to emergency services.
- Rural Areas:** Could extend up to 1-2 hours given the distance and road conditions.

## 12. Overall Summary of Risk Assessment

The route is largely safe for heavy vehicle travel, with a few concerns primarily relating to weather, road conditions in rural areas, and peak-hour traffic congestions. Ensuring adherence to local regulations and preparing contingency plans for communication and emergency rerouting will enhance safety. Awareness of environmental and weather-related challenges will aid in smooth transit, with adequate risk mitigation measures in place for hazardous materials.

By understanding and preparing for the highlighted factors, truck drivers can minimize risks, ensuring both safety and regulatory compliance during transit.

### Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Roundabout | High       | 11.38395, 77.89480 | 15 KM/Hr    |
| 1  | Roundabout | High       | 11.04637, 77.73761 | 15 KM/Hr    |
| 2  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 3  | Turn       | High       | 11.44022, 77.87546 | 15 KM/Hr    |
| 4  | Turn       | Medium     | 11.39976, 77.88382 | 30 KM/Hr    |
| 5  | Turn       | Medium     | 11.39690, 77.88852 | 30 KM/Hr    |
| 6  | Turn       | High       | 11.37868, 77.89498 | 15 KM/Hr    |
| 7  | Turn       | Medium     | 11.37857, 77.89473 | 30 KM/Hr    |
| 8  | Turn       | High       | 11.36616, 77.88885 | 15 KM/Hr    |
| 9  | Turn       | Medium     | 11.34067, 77.86347 | 30 KM/Hr    |
| 10 | Turn       | Medium     | 11.33854, 77.86216 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 11 | Turn       | Medium     | 11.33778, 77.86236 | 30 KM/Hr    |
| 12 | Turn       | Medium     | 11.30952, 77.77587 | 30 KM/Hr    |
| 13 | Turn       | Medium     | 11.30973, 77.76712 | 30 KM/Hr    |
| 14 | Turn       | High       | 11.29908, 77.75818 | 15 KM/Hr    |
| 15 | Turn       | Medium     | 11.18935, 77.77078 | 30 KM/Hr    |
| 16 | Turn       | High       | 11.16434, 77.77360 | 15 KM/Hr    |
| 17 | Turn       | Medium     | 11.16375, 77.77424 | 30 KM/Hr    |
| 18 | Turn       | Medium     | 11.13659, 77.76519 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.04614, 77.73704 | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.04505, 77.73623 | 30 KM/Hr    |
| 21 | Turn       | High       | 10.99023, 77.72506 | 15 KM/Hr    |
| 22 | Turn       | High       | 10.98970, 77.72385 | 15 KM/Hr    |
| 23 | Blind Spot | Blind Spot | 10.94597, 77.71319 | 10 KM/Hr    |
| 24 | Turn       | High       | 10.94563, 77.71574 | 15 KM/Hr    |
| 25 | Turn       | Medium     | 10.92228, 77.72392 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 10.86278, 77.72143 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 10.86158, 77.71764 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 10.86118, 77.71726 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 10.85613, 77.71791 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 10.85462, 77.72064 | 30 KM/Hr    |
| 31 | Turn       | Medium     | 10.84875, 77.72076 | 30 KM/Hr    |
| 32 | Turn       | Medium     | 10.82285, 77.72027 | 30 KM/Hr    |
| 33 | Turn       | High       | 10.81280, 77.71753 | 15 KM/Hr    |
| 34 | Turn       | Medium     | 10.81256, 77.72035 | 30 KM/Hr    |
| 35 | Turn       | Medium     | 10.80911, 77.72390 | 30 KM/Hr    |
| 36 | Turn       | High       | 10.80865, 77.72416 | 15 KM/Hr    |
| 37 | Turn       | Medium     | 10.80868, 77.72432 | 30 KM/Hr    |
| 38 | Turn       | High       | 10.80774, 77.72667 | 15 KM/Hr    |
| 39 | Turn       | High       | 10.80125, 77.72786 | 15 KM/Hr    |

## Emergency Locations

|    | type     | name                             | coordinates            | speed_limit | risk_level |
|----|----------|----------------------------------|------------------------|-------------|------------|
| 0  | hospital | Tiruchengode, Goverment Hospital | 11.3903328, 77.8920627 | 30 km/h     | Medium     |
| 1  | hospital | SPM Medical Centre,Tiruchengode  | 11.3881331, 77.8931963 | 30 km/h     | Medium     |
| 5  | clinic   | Kongu Nursing Home               | 11.3783065, 77.8961134 | 30 km/h     | Medium     |
| 6  | hospital | T.C.A Hospital Tiruchengode      | 11.3791885, 77.8965774 | 30 km/h     | Medium     |
| 7  | hospital | Soorya Multispecialty Hospital   | 11.3786429, 77.8931912 | 30 km/h     | Medium     |
| 8  | hospital | Tiruchengode Government Hospital | 11.37645, 77.89426     | 30 km/h     | Medium     |
| 9  | hospital | Tirukumaran Hospitals            | 11.3782118, 77.8914933 | 30 km/h     | Medium     |
| 10 | hospital | Krishna Hospital, Namakkal       | 11.3754811, 77.8931817 | 30 km/h     | Medium     |
| 13 | hospital | Keerthana Hospital               | 11.2365339, 77.7772287 | 30 km/h     | Medium     |
| 14 | hospital | Nataraj Nursing Home             | 11.2351682, 77.7785956 | 30 km/h     | Medium     |
| 15 | hospital | Government Hospital, Elumathur   | 11.18533, 77.772496    | 30 km/h     | Medium     |
| 16 | hospital | Malar Hospital, Muthur           | 11.0438049, 77.7361156 | 30 km/h     | Medium     |
| 17 | hospital | SIBI Hospital, Vellakoil         | 10.9569141, 77.7160216 | 30 km/h     | Medium     |
| 18 | hospital | KG Hospital, Vellakoil           | 10.9511, 77.714116     | 30 km/h     | Medium     |

## Crowded Spots

|    | type        | name                                | coordinates            | speed_limit | risk_level |
|----|-------------|-------------------------------------|------------------------|-------------|------------|
| 2  | school      | அரசு ஆண்கள் மேல்நிலைப் பள்ளி        | 11.3850439, 77.8948279 | 30 km/h     | Medium     |
| 3  | school      | அரசு பெண்கள் மேல்நிலைப் பள்ளி       | 11.3844247, 77.8949053 | 30 km/h     | Medium     |
| 4  | marketplace | திருச்செங்கோடு தினசரி காய்கறி சந்தை | 11.3833608, 77.8970145 | 30 km/h     | Medium     |
| 11 | school      | KSR Educational institution         | 11.3772013, 77.8908807 | 30 km/h     | Medium     |
| 12 | school      | MDV School                          | 11.3719843, 77.8915244 | 30 km/h     | Medium     |

## Route Photos of Risky Spots





**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.38395, 77.89480



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.04637, 77.73761





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44022, 77.87546



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.39976, 77.88382



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.39690, 77.88852





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.37868, 77.89498



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.37857, 77.89473





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.36616, 77.88885



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.34067, 77.86347



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.33854, 77.86216





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.33778, 77.86236



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.30952, 77.77587





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.30973, 77.76712**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.29908, 77.75818**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.18935, 77.77078**





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.16434, 77.77360



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.16375, 77.77424





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.13659, 77.76519



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.04614, 77.73704



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.04505, 77.73623





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 10.99023, 77.72506



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 10.98970, 77.72385





**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 10.94597, 77.71319



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.94563, 77.71574



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.92228, 77.72392





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 10.86278, 77.72143



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 10.86158, 77.71764





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.86118, 77.71726



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.85613, 77.71791



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.85462, 77.72064





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 10.84875, 77.72076



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 10.82285, 77.72027

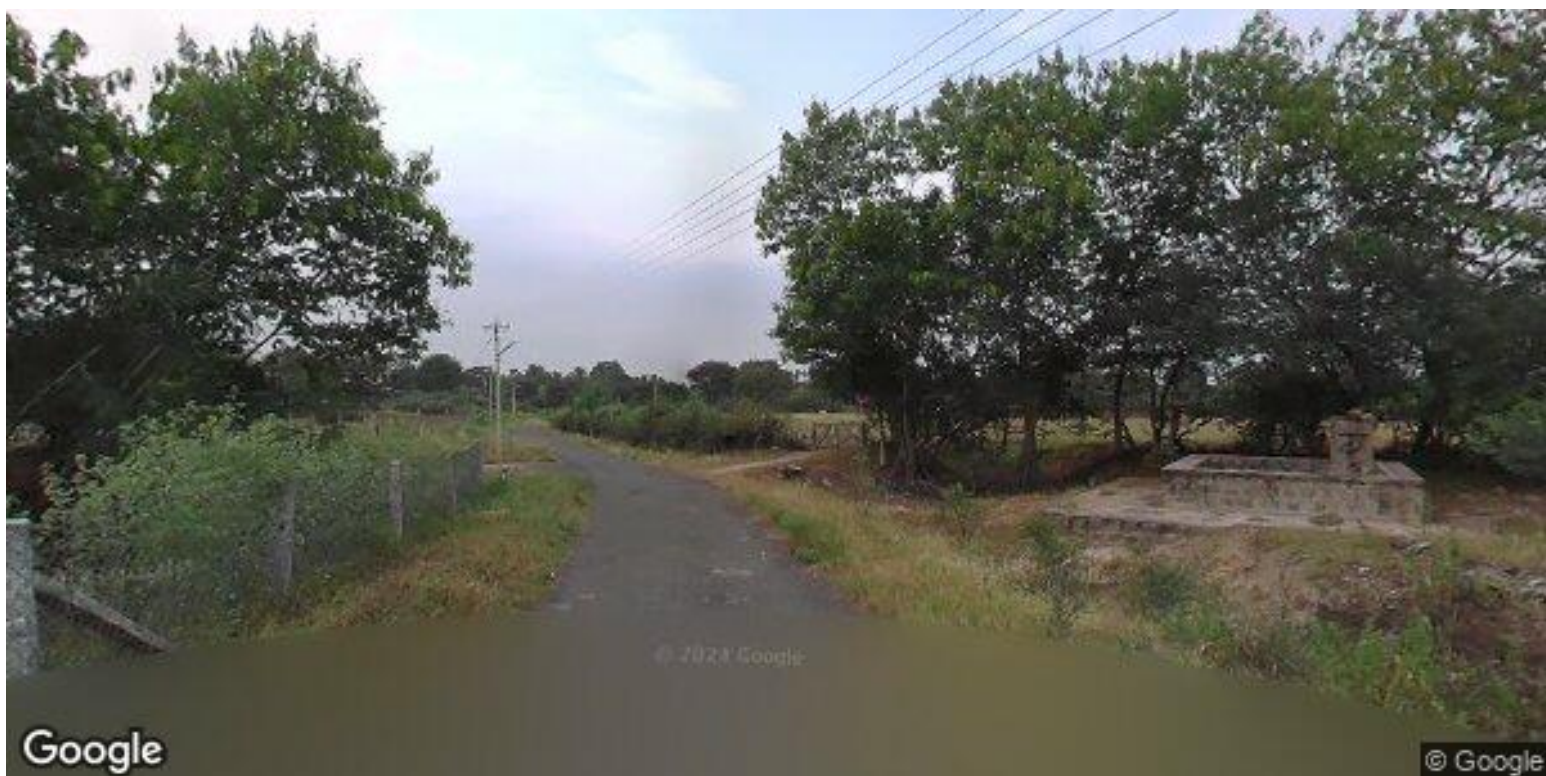




**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.81280, 77.71753

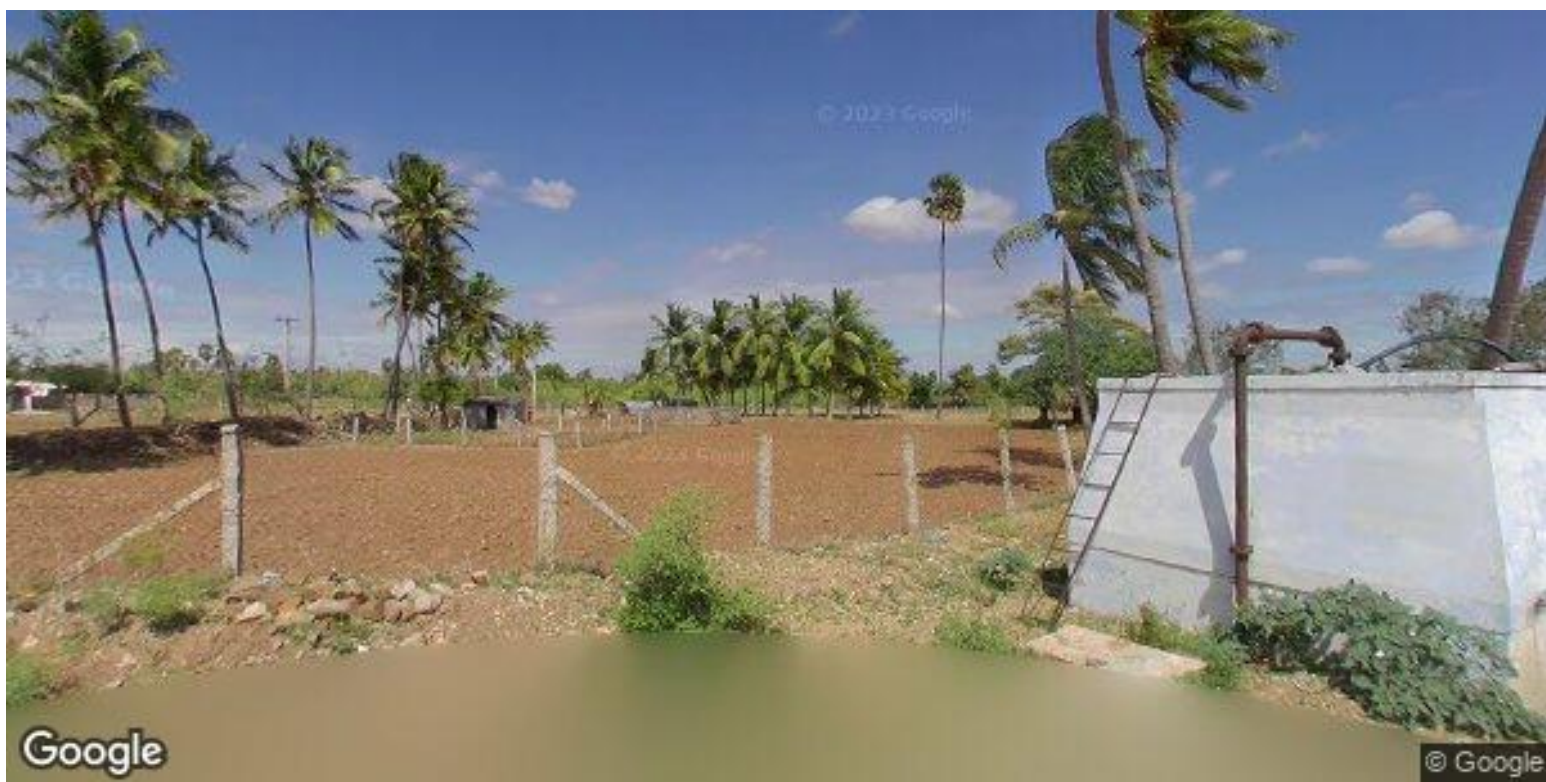


**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.81256, 77.72035



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 10.80911, 77.72390





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 10.80865, 77.72416

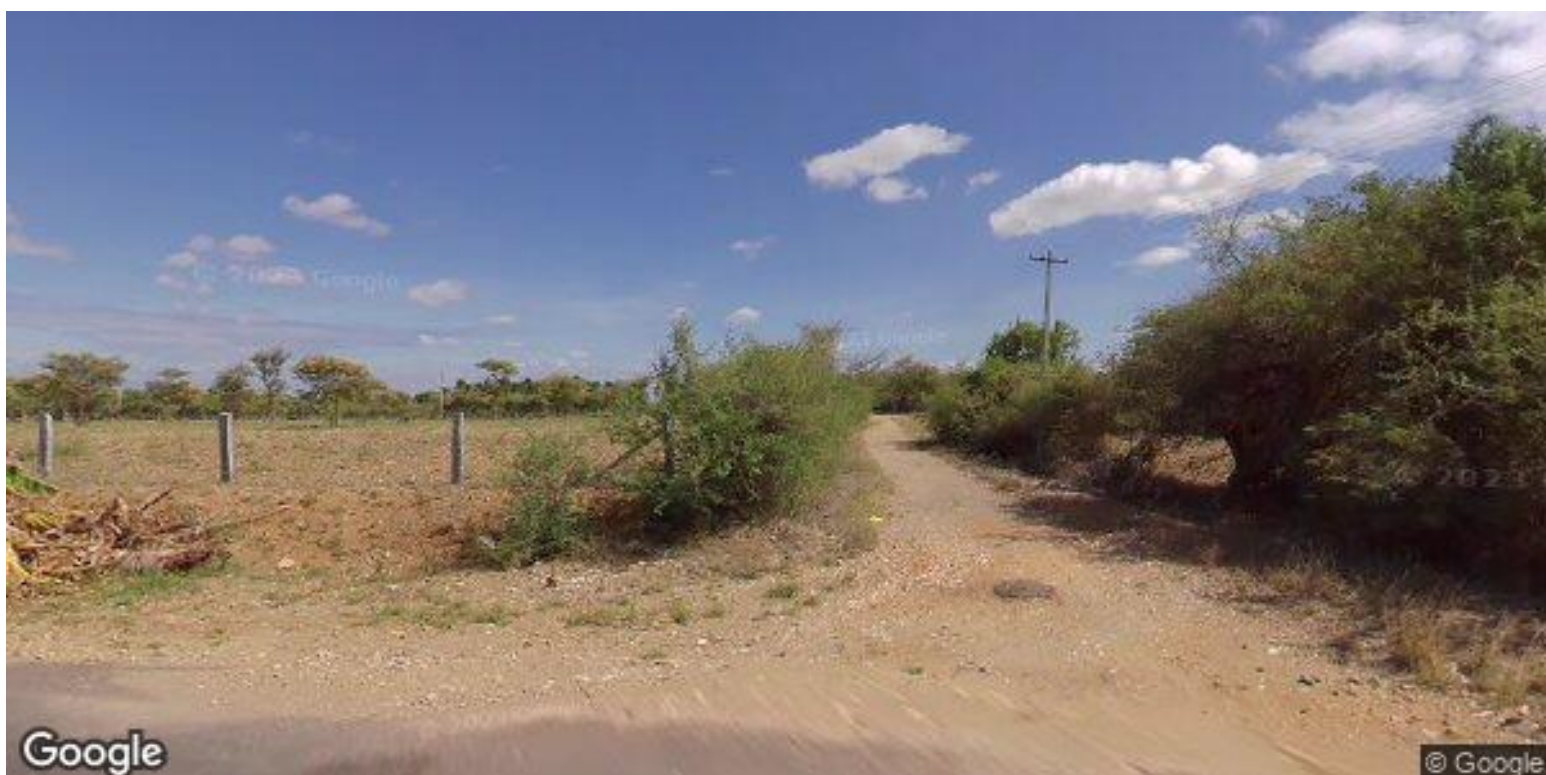


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 10.80868, 77.72432





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 10.80774, 77.72667



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 10.80125, 77.72786

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